

Heron of Alexandria in the Renaissance



Born and Died: First Century A.D.

Sara Taglialagamba

Lamporecchio, Nuova Fondazione Rossana e
Carlo Pedretti, Università Carlo Bo, Urbino, Italy

Abstract

Heron of Alexandria, mathematician, physicist, and engineer of Greek origin, was considered the greatest experimenter of antiquity and the leading figure of the Roman and Hellenistic scientific tradition. He taught at Alexandria's *Musaeum* and he wrote several treatises on Mathematics, Geometry, and Engineering. Most of his books have been lost, but fortunately some of them were preserved mainly in Arabic manuscripts. Among his outstanding inventions, one could mention the Aeolipile, the first steam turbine, automated machines for temples and theatres, surveying instruments, and military machines and weapons.

Biography, Heritage and Rupture with the Tradition, Innovative and Original Aspects, Impact and Legacy

The amount of information that has reached us about one of the greatest mathematicians,

engineers, physicists, and inventors of antiquity is quite small. Since his name was very common, to distinguish him from other namesake authors such as Proclus and Damiano called him μηχανικός, i.e., “Mechanic” or “Engineer.” As well as praising him for his mechanical achievements, in his *Collectiones* (third century A.D.), Pappus referred to him as Ἀλεξανδρεὺς, confirming some of the geographic mentions contained in his works, such as Alexandria in Egypt, which was part of the Roman Empire at that time and where Heron taught at the prestigious museum. Many inconsistencies make it difficult to determine the exact dates and chronology of his life. In his works he referred to Archimedes († 212 B.C.), Apollonius († 150 B.C.), and to the chords of the circle that can be traced back to Hipparchus of Nicaea († 125 B.C.) and Posidonius († 51 A.D.). Although Heron's debt toward his great predecessor Ctesibius is undisputed, the notes of an anonymous, tenth century Byzantine author lead to the erroneous belief that Ctesibius – who according to the assertions of Athenaeus, Diodorus Siculus, and Pliny the Elder may have lived in the third century B.C. – was his lecturer. Heron mentioned Philo of Alexandria († 50) as if he were still alive and his contemporary. A trustworthy fact can be drawn from the analyses of Chap. 35 of *Dioptra* in which Heron calculated the distance between Rome and Alexandria using various local times and referring to a lunar eclipse that can be identified in the one that occurred on March 13th, 62 A.D. Despite

many objections, this date seems to be the most reliable one and assigns Heron to the first century A.D. A keen writer, an expert technologist, and a strong advocate of the need for all-round education including both theory and practice, in his works Heron tackled various fields of scientific knowledge, achieving important results in mechanics, mathematics, geometry, physics, optics, geodesy, hydraulics, and pneumatics. He always managed to strike a balance between his theoretical interest for complex problems, especially mathematical ones, and their solution by means of technical-practical applications. Since his treatises were very useful, his manuscripts (which were later printed) became exceptionally successful but were also continuously modified and added to by subsequent authors who are often difficult to identify. Heron's works can be divided into two categories: theoretical works, mainly regarding mathematics and geometry (*Metrica*, *Definitiones*, *Geometrica*, *Geodaesia*, *Stereometrica*, *De Mensuris*, *Liber Geeponicus*, *Commentary on Euclid's elements*) and technological writings that describe the constructive, practical application of the solution of mathematical problems. The latter can be further divided into two sub-categories: technological-theoretical works, i.e., those regarding mechanics and optics (*Mechanica*, *Pneumatica*, *Automata*, *Barulcos*, *Dioptra*, and *Catoptrica*) and technological-practical writings concerning the construction of machines and war devices (*On water clocks*, *Belopoeica*, and *Cheirobalistra*). Heron suggested brilliant ways of solving measurement problems; he correctly defined the laws of reflection, invented a formula to calculate the area of the triangle from its side lengths and described a method for computing square and cube roots. He invented mechanical toys designed to astonish onlookers. His outstanding inventions include *automata*, machines for temples and theatres. He also studied the properties of siphons that were essential to the construction of amazing water features. Among his most important inventions were the Heron's fountain; the odometer; the *Aeolipile*, the first steam turbine, and the *Cheirobalistra*, a powerful catapult.

Heron's writings came as fragments, appearing as lecture notes for courses in mathematics, mechanics, physics, and pneumatics, but they were the first Greek works to be translated. The keen interest in technology and machines in the Renaissance led Heron to great popularity, especially thanks to *Pneumatica* but also as a result of *Automata*. However, Heron's masterpiece is his treatise *Mechanica* consisting of two books in Arabic: the first was written for architects while the second describes five simple machines – the lever, the winch, the pulley, the screw, and the wedge – all of which work by applying the law of the lever. Known also by Leonardo da Vinci, his writings appeared in Leon Battista Alberti's *Ludi Matematici* (1450–1452) and in Giorgio Valla's encyclopaedia (1501), followed by complete translations into Latin by Federico Commandino in 1575 (*Heronis Alex. Spiritualium liber*) and subsequently a translation and a printed edition by Giovanni Battista Aleotti (*Gli Artificiosi et Curiosi Moti Spirituali di Herone Alessandrino*, Ferrara 1589 and Bologna 1647) in which the translator incorporated some of his own ideas. Other versions by Commandino's pupils in Urbino followed soon after, such as those by Alessandro Giorgi, Guidobaldo del Monte, and Bernardino Baldi. *Le due regole della prospettiva pratica* by M. Iacopo Barozzi da Vignola, published posthumously in 1583 in Rome, with a commentary by Dominican friar Egnazio Danti, attests to the fact that his mechanisms for the theatre were well-known and used. In particular Hero's most important contributions to hydraulics is probably his correct understanding of the relationship between discharge (i.e., equal to volume per time unit), cross-sectional area, and velocity. This important finding remained unnoticed until Renaissance with scientists Leonardo da Vinci and Benedetto Castelli.

Cross-References

- [Hydraulics](#)
- [Pneumatics](#)

References

Primary Literature and Secondary Literature

Bibliography Heron of Alexandria

- Canali, Ferruccio. 1999. Per un primo corpus di testi di teoria architettonica nell'età antica: Vitruvio Pollione, Erone Alessandrino, Epafrodito e Vitruvio Rufo. In *Saggi di letteratura architettonica*, 85–103. Florence: Pontecorboli.
- Cantor, M. 1908. *Vorlesungen über Geschichte der Mathematik I*. Leipzig.
- Derry, Thomas Kingston, and Trevor Williams. 1993. *A short history of technology. From the earliest times to A.D. 1900*. New York: Dover Publications.
- Drachmann, Aage G. 1948. *Ktesibios, Philon, and Heron: A study in ancient pneumatics*. Copenhagen: E. Munksgaard.
- Drachmann, Aage G. 1963. *The mechanical technology of Greek and Roman antiquity*. Copenhagen: E. Munksgaard.
- Drachmann, A.G., and M.S. Mahoney. 1970–1990. *Biography in dictionary of scientific biography*. New York: Charles Scribner & Sons.
- Gatto, Romano. 2012. La «Meccanica» di Erone nel Rinascimento. In *Scienze e rappresentazioni: Saggi in onore di Pierre Souffrin*, 151–172. Florence: Olschki.
- Giardina, G. Giovanna. 2003. *Erone di Alessandria: Le radici filosofico-matematiche della tecnologia applicata: «Definitiones»: Testo, traduzione e commento*. Catania: CUECM.
- Heath, Thomas L. 1931. *A history of Greek mathematics, 2 vols.* Oxford, UK: Oxford University Press.
- Heiberg, Johannes Ludwig. 1912. *Heronis Alexandrini Opera quae supersunt omnia*. Leipzig: Teubner Series.
- Heron of Alexandria. Biography in encyclopaedia Britannica. Web: <http://www.britannica.com/eb/article-9040189/Heron-of-Alexandria>
- Hooykaas, Reijer. 1953. La théorie corpusculaire de Léonard de Vinci. In *Léonard de Vinci et l'expérience scientifique au XVIe siècle*, 163–169. Paris: Centre National de la Recherche Scientifique: Universitaires de France.
- Landels, Jhon. 2000. *Engineering in the ancient world*. rev. ed. Berkeley/Los Angeles: University of California Press.
- Lazos, Christos D. 1999. *The adventure of technology in ancient Greece*. Athens: Aiolos.
- Lewis, Michael J.T. 2001. *Surveying instruments of Greece and Rome*. Cambridge, UK: Cambridge University Press.
- Lloyd, Geoffrey E. 1973. *Greek science after Aristotle*. New York: W.W. Norton & Co.
- Lloyd, Geoffrey E. 1991. *Methods and problems in Greek science*. Oxford, UK: Oxford University Press.
- Marsden, Eric William. 1969. *Greek and Roman artillery: Historical development*. Oxford, UK: Clarendon Press.
- Marsden, Eric William. 1971. *Greek and Roman artillery: Technical treatises*. Oxford, UK: Clarendon Press.
- Neugebauer, Otto. 1938. Über eine Methode zur Distanzbestimmung Alexandria – Rom bei Heron. *Kongelige Danske Videnskabernes Selskabs Skrifter* 26: 21–24.
- Neugebauer, Otto. 1975. *A history of ancient mathematical astronomy*. Berlin/Heidelberg/New York: Springer.
- O'Connor, J.J., and E.F. Robertson. 1999. Heron biography. *The MacTutor history of mathematics*. <http://www-history.mcs.st-andrews.ac.uk/history/Biographies/Heron.html>
- Papadopoulos, Evangelos. 2007. Heron of Alexandria (c. 10–85 AD). In *Distinguished figures in mechanism and machine science*, vol. 1, 217–245. Cham: Springer.
- Schmidt, W. 1899. *Heronis Alexandrini Opera quae supersunt omnia*, vol. I and supplement. Leipzig: Teubner Series.
- Trabucco, Oreste. 2010. “L'opere stupende dell'arti più ingegnose”: La recezione degli Pneumatiká di Erone Alessandrino nella cultura italiana del Cinquecento. Florence: Olschki.
- Tuplin, C.T., and Tracey Elizabeth Rihll. 2002. *Science and mathematics in ancient Greek culture*. Oxford, UK: Oxford University Press.
- Woodcroft, B. 1851. *The pneumatics of hero of Alexandria*. Translated and edited from the original Greek. London: Taylor & Maberly.